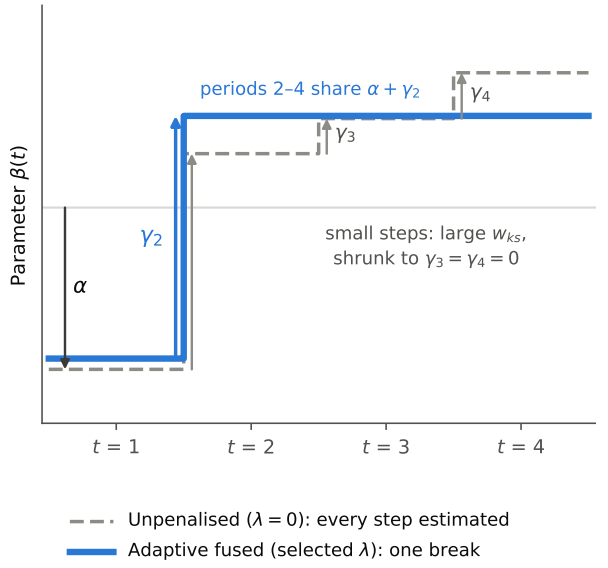
(a) Level and step variables for an indicator $x = 1$

Period	t	x	$x D_2$	$x D_3$	$x D_4$	**Parameter** $\beta(t)$
2016–17	1	1	0	0	0	α
2018–19	2	1	1	0	0	$\alpha + \gamma_2$
2020–21	3	1	1	1	0	$\alpha + \gamma_2 + \gamma_3$
2022–23	4	1	1	1	1	$\alpha + \gamma_2 + \gamma_3 + \gamma_4$

$D_s = 1[t \geq s]$: step indicator, so $\gamma_s = \beta(s) - \beta(s-1)$, the change between adjacent periods. The adaptive penalty shrinks small steps to exactly zero; with $\gamma_3 = \gamma_4 = 0$, periods 2–4 share one parameter (panel b).

(b) The steps of (a) before and after the penalty